Buoyancy

We know that gravity _____ objects _____. The force is known as _____.

But underwater there's another, upward force, called _____.

If the weight of an object is _____ than the buoyant force, the object will _____.

If the buoyant force is greater than the _____, the object will _____.

The amount of buoyant force is determined by the weight of water an object moves out of the way - known as the _____ _____.

Buoyant Force:

An _____ acting force exerted by a _____

Opposes an object's _____

The greater the weight of the displaced water, the greater the buoyant force.

Whether an object floats or sinks depends on the _____ of the _____ in comparison to the density of the _____ it displaces.

Density = amount of _____ in a set _____

If the density of the object is _____ than that of the water it's displaced, its weight will be greater too and it will _____.

If it's _____ dense, it will float.

The effects of changing the density can be seen in the ease with which people float on the salty water of the Dead Sea.

The salt increases the _____ of the water, so the buoyant force is much larger.

Buoyancy determined by:

_____ of displaced water and object

_____ of displaced water and object

These _____ help explain why heavy metal ships float.

Firstly, they're ____ - they displace a huge amount of water, creating a very strong buoyant force.

And secondly, they are _____ - combined, the _____ _____ and ___ in the hull are less dense than water, so the weight is less than the buoyant force.

If ships were made of solid metal, they would _____!